

Breakout Swing Scanner Pro Tutorial

Installation, configuration, Pine Screener setup, alerts, and operating workflow.

Script: Breakout Swing Scanner Pro (Daily + Intraday ORB)

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Use case: TradingView Pine Script® v6 indicator for chart, watchlist, and Pine Screener workflows.

Educational material only. This document is not financial, investment, legal, or tax advice.

1. What You Received

The package contains an upgraded Pine Script® v6 indicator named **Breakout Swing Scanner Pro**. It keeps the original scanner objective—daily Donchian breakouts and intraday Opening Range Breakouts—while tightening signal hygiene, screener compatibility, and operating instructions.

Deliverables: improved `.pine` file, this tutorial PDF, a methodology whitepaper PDF, and a zipped bundle.

2. Major Improvements in the Pro Version

Area	Improvement	Why it matters
Intraday daily context	Uses last-confirmed daily data with <code>lookahead=barmerge.lookahead_on</code> and prior-bar expressions.	Reduces higher-timeframe repaint risk on intraday charts.
Session handling	When session filtering is off, ORB and VWAP now evaluate the active chart day instead of silently continuing to rely on the regular-session test.	Makes all-hours workflows more consistent.
Opening range sizing	Uses a ceiling calculation for ORB bars.	Avoids undercounting the opening range on coarser intraday intervals.
Benchmark input	Uses <code>input.string()</code> for the benchmark symbol rather than <code>input.symbol()</code> .	Maintains Pine Screener compatibility while allowing users to change the benchmark.
Risk controls	Adds ATR stop-risk percentage and an optional max-risk filter.	Helps reject signals whose volatility-defined stop is too wide.
Alerts	Keeps <code>alertcondition()</code> triggers and adds optional detailed <code>alert()</code> messages.	Supports both screener/watchlist alert workflows and richer webhook-style messages.
Usability	Adds diagnostic plots and a dashboard.	Makes it easier to understand why a signal is or is not ready.

3. Installation

1. Open TradingView and go to **Supercharts → Pine Editor**.

2. Open the downloaded file `Breakout_Swing_Scanner_Pro_v6.pine`.
3. Copy the full script into the Pine Editor.
4. Click **Save**, then **Add to chart**.
5. For Pine Screener use, add the script to your favorites so it appears in the Screener indicator dropdown.

Important: The script is an indicator, not a strategy. It does not simulate fills, slippage, partial exits, commissions, or order execution.

4. Recommended Starting Presets

Workflow	Timeframe	Run mode	Intraday mode	Suggested filters
Daily swing breakout	1D	Auto or Daily only	Not used	Trend on, Volume on, ADX optional, RS optional
Opening range day trade scan	5, 15, or 30 min	Auto or Intraday only	Opening Range	VWAP on, Volume on, daily trend on
Higher-conviction intraday confirmation	5, 15, or 30 min	Intraday only	Both (confirm)	VWAP on, Volume on, Market regime optional
Continuation after daily breakout level	15, 30, 60 min	Intraday only	Daily Donchian	Trend on, Volume on, optional max ATR risk

5. Input Guide

Core

- **Run mode:** *Auto* uses intraday logic on intraday charts and daily logic otherwise. *Daily only* and *Intraday only* force a specific mode.
- **Enable long/short signals:** Lets you build long-only, short-only, or two-sided scans.
- **Intraday signal throttling:** *One per session/day* prevents repeated intraday signals in the same direction; *Every qualifying bar* allows repeats if `freshOnly` is off.

Breakout

- **Donchian lookback:** Number of prior bars/days used to form breakout levels. The current bar is excluded from the level.
- **Fresh breakout only:** Requires a crossover/crossunder on the current confirmed bar.
- **Use Close for breakout:** More conservative. Turning it off uses high/low for more aggressive detection.
- **Opening range minutes:** Defines the ORB window. The script rounds up to the required number of chart bars.

Session

For equities, the default regular session is `0930-1600` in the exchange timezone. For futures, crypto, forex, or non-U.S. listings, update the session string or turn session filtering off.

On large intraday bars, session boundaries can fall inside a bar. Use intervals that align with your session whenever possible; otherwise, the first in-session bar may not be exactly at the session open.

Filters

- **Volume:** Requires current volume to exceed a moving average by the selected multiplier.
- **EMA trend:** Long signals require fast EMA above slow EMA and price above slow EMA; shorts require the inverse.
- **VWAP:** Intraday long signals require price above session VWAP; shorts require price below session VWAP.
- **ADX:** Optional trend-strength gate.
- **Relative strength:** Compares symbol/benchmark ratio against its moving average.
- **Market regime:** Uses the prior confirmed daily benchmark close versus its slow EMA.

Risk

The suggested stop is `close ± ATR × multiplier`. For long signals, the stop is below close; for short signals, it is above close. The optional max ATR stop-distance filter rejects breakouts when the volatility-defined stop is too wide for your risk model.

6. Pine Screener Setup

1. Open **Products** → **Screeners** → **Pine**.
2. Select the watchlist you want to scan.
3. Select the saved/favorited **Breakout Swing Scanner Pro** indicator.
4. Choose the Screener timeframe. For daily scans, use **1D**. For ORB scans, use a supported intraday interval such as **5**, **15**, or **30** minutes.
5. Add filters such as `LongSignal (1/0) = 1` or `ShortSignal (1/0) = 1`.
6. Sort by `VolRatio_or_RelVol`, `ADX`, `RSRatio vs Benchmark`, or `ATRStopRiskPct` depending on your process.

The first ten `plot()` calls are reserved as Screener-facing columns. Additional diagnostic plots are also available but may need to be manually enabled as columns.

7. Column Reference

Column	Meaning	Typical use
LongSignal (1/0)	1 when all long conditions pass.	Filter equals 1.
ShortSignal (1/0)	1 when all short conditions pass.	Filter equals 1.
ActiveLevelLong	Current long breakout level.	Compare with price or alert output.
ActiveLevelShort	Current short breakdown level.	Compare with price or alert output.
VolRatio_or_RelVol	Volume divided by its moving average.	Sort for participation.
ADX	Current timeframe ADX.	Trend-strength ranking.
VWAPDevPct (intraday)	Percent distance from session VWAP.	Intraday extension/positioning.
RSRatio vs Benchmark	Relative-strength ratio divided by its moving average.	Above 1 favors long RS; below 1 favors short RS.
SuggestedStopPrice	ATR stop price when a signal fires.	Initial risk planning.
ModeCode	1 = Daily, 2 = Intraday.	Confirm active mode in Screener.

8. Alert Setup

The script exposes separate **Breakout Long** and **Breakout Short** alert conditions. Create them from the TradingView alert dialog by selecting the script and the desired condition. If you enable **Also trigger detailed alert() messages**, create an alert using **Any alert() function call** to receive richer text including mode, close, level, and stop.

When you edit a script after creating an alert, recreate the alert so the alert server uses the current script snapshot and inputs.

9. Troubleshooting

Issue	Likely cause	Fix
No signals in Screener	Filters too strict; wrong mode; unsupported timeframe; warmup bars missing.	Turn off optional filters one at a time, use Auto mode, and test on 1D or 5/15/30 minute intervals.
ORB level starts late	Chart interval does not align with session start.	Use a smaller or aligned timeframe.
Benchmark filters block everything	Benchmark symbol unavailable for the market/session.	Disable RS/market regime or set a valid benchmark string.
Stop is na	ATR warmup not complete or data unavailable.	Use more history or lower ATR length.
Signals differ from a live intrabar move	Script waits for confirmed bars.	This is intentional; use lower timeframes for earlier confirmed bars.

10. Practical Workflow

1. Build a liquid watchlist first. The scanner is only as useful as the universe it scans.
2. Run a broad scan with minimal filters to see signal frequency.
3. Add trend, volume, VWAP, RS, and risk filters gradually.
4. Review chart context manually before acting on any signal.
5. Record signal outcomes and refine settings by market, timeframe, and symbol type.

11. Compliance and Risk Note

This tool is for research and monitoring. It does not guarantee profitable outcomes, and breakout systems are vulnerable to failed breakouts, gap risk, liquidity events, data issues, and market regime changes. Always validate parameters with your own historical research and risk controls.

References

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